THABRE ALAM SHAHUL HAMEED A

Electronics & Communication Engineer | Embedded Systems, VLSI, Automotive Electronics & Data Analytics
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PROFESSIONAL SUMMARY

Final-year ECE student (CGPA 8.00/10) with hands-on experience in Embedded C, Verilog HDL and FPGA design (Xilinx Vivado), automotive tools (CANoe, TSMaster, ADAS training) and image-processing-based embedded systems. Also skilled in Python, SQL and Power BI analytics, with a full-stack project built using Next.js, TypeScript and Supabase. Seeking core engineering and technical roles in electronics, embedded, automotive and IT.

EDUCATION

B.E. Electronics & Communication Engineering — University College of Engineering, Tiruchirappalli *2023 – 2027 | CGPA: 8.00/10*
Higher Secondary (12th) & SSLC (10th) — St. Joseph's College Hr. Sec. School, Trichy *12th: 86.5%*

TECHNICAL SKILLS

Languages: C, Embedded C, Python, TypeScript, Verilog HDL, MATLAB, SQL, HTML
VLSI & Embedded Tools: Xilinx Vivado, Xilinx ISE, ModelSim, Keil uVision, Proteus VSM, GNU Radio, CANoe, TSMaster
Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Microsoft Excel, Power BI, MySQL, SQL Server
Web & Dev Tools: Flask, Supabase, REST APIs, Gmail API, Google OAuth 2.0, Twilio, VS Code, GitHub, Anaconda

PROJECTS

FPGA Implementation of LDPC Encoder and Decoder for Satellite Communication *Vivado, Verilog HDL, Nexys 4 DDR*
- Designed and implemented an LDPC encoder and decoder on an Artix-7 FPGA for reliable satellite communication; performed synthesis, implementation and hardware validation in Vivado.

FIR Low Pass Filter using Distributed Arithmetic *Xilinx Vivado, Verilog HDL*
- Designed a FIR Low Pass Filter using Distributed Arithmetic; verified via simulation and timing analysis.

Vehicle Number Plate Detection System *Embedded C, Keil uVision, Proteus*
- Built a system to detect and recognize vehicle registration numbers, integrating image processing with embedded hardware.

Email Keyword Call Alert System *Next.js, TypeScript, Gmail API, Pub/Sub, Supabase, Twilio*
- Built a web app that monitors Gmail for user-defined keywords and triggers automated Twilio phone call alerts on matching emails, using Google OAuth 2.0, Gmail API and Google Pub/Sub event notifications.
- Implemented keyword detection, email content extraction, alert logging and duplicate-alert prevention (TypeScript, Supabase); built a dashboard for keywords, voice settings, call templates and call history.

AI Job Market Analysis Dashboard *MySQL, SQL, Power BI, DAX*
- Cleaned data with SQL and built an interactive Power BI dashboard on salary trends, hiring patterns and skills.

Cryptocurrency Price Prediction using Live Market Data *Python, REST API, Scikit-learn*
- Collected live API data, engineered features with Pandas/NumPy, and built ML models to predict price movement.

INTERNSHIP EXPERIENCE

Data Analytics Intern — VDart Academy, Tiruchirappalli *Jul – Aug 2025 & Jun – Jul 2026*
- Cleaned and analysed datasets using Excel, Power BI, SQL and Python (Pandas, NumPy, Matplotlib, Seaborn).

India Space Lab Summer Internship & Technical Training (Online) *Jun – Jul 2026*
- Completed training in Advanced Drone Technology, CanSat, CubeSat, Rocketry, Remote Sensing & GIS, and Disaster Management.

Digital VLSI Design Intern — National Institute of Technology, Tiruchirappalli *Jan – Feb 2026*
- Designed and simulated digital circuits in Verilog HDL and Vivado; implemented a Distributed Arithmetic FIR filter.

Analog & Digital Circuit Design Intern — National Institute of Technology, Tiruchirappalli *Jan – Feb 2025*
- Designed and simulated analog and digital circuits using Proteus VSM; analysed performance and validated results.

CERTIFICATIONS

- Data Science & Machine Learning Fundamentals — ROOT-IT Learning Centre, Trichy (6 months, Grade A; Sept 2026)
- BIT2CHIP: Making RTL Real on FPGA Using Vivado — PSG Institute of Technology, Coimbatore (Feb 2026)
- IoT and Robotics Workshop — Sona College of Technology, Salem (Mar 2026)
- SNAP ON CHIP Image Processing Workshop (sponsored by Qualcomm) — NIT Tiruchirappalli (Feb 2025)
- NPTEL — Introduction to Machine Learning (Elite, 61%) | ADAS Advanced Level 1 & 2 — Naan Mudhalvan Scheme

ACHIEVEMENTS

- Best Paper Presenter Award, International Multidisciplinary Conference MCKSHPS 2026, for "Smart Education Technologies using Recommendation System" (published in proceedings).

LANGUAGES & STRENGTHS

Tamil (Native), English (Proficient) | Team player, quick learner, adaptable, public speaker, punctual